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Impact of small-molecule adsorbates on the morphology of PuO₂ nanoparticles from first-principles modelling

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Abstract

The safe management of legacy civil plutonium stockpiles is among the most difficult challenges facing the nuclear industry. While geological disposal facilities (GDFs) are seen as the optimal solution, anticipating the evolution of waste forms over the lifetime of the GDF forms a critical part of the safety case. In the typical storage form of PuO₂ powders, the chemical reactivity of Pu is determined by the exposed crystal facets and surface speciation, which are a complex function of temperature and the partial pressures of oxygen and small-molecule adsorbates. In this work, we use a first-principles modelling approach to develop a predictive thermodynamic model for the impact of the ubiquitous environmental compounds H₂O, CO₂ and H₂O₂ on the equilibrium particle morphology and surface speciation of stoichiometric and oxygen-deficient PuO₂. We find that the presence of multiple adsorbates can lead to both synergistic and antagonistic interactions, with significant impacts on the energetically-accessible nanoparticle morphologies and the exposure of the major {100}, {110} and {111} facets. Our model provides important reference data for the impact of environmental conditions on the surface chemistry of PuO₂, and can be systematically enhanced to account for other variables including additional adsorbates, solvation, and surface coverage.

1. Introduction

The UK holds the largest quantity of civil plutonium in the world, an estimated 140 tonnes, which is currently stored at the Sellafield site [1]. Many other countries, including the United States, France and Japan, also hold significant quantities of Pu in interim storage or contained in wastes, weapons or nuclear fuels [2].

The proliferation risk and long decay half-life makes managing Pu stockpiles both challenging and expensive. The UK government recently committed to immobilising the Pu stored at Sellafield for disposal in a geological disposal facility (GDF) [3]. The Pu will need to be stored for millennia, and the multiple engineered barriers created to contain it will eventually fail and bring the stored Pu into contact with groundwater and/or other non-inert environments. Understanding the chemical processes that could take place, and whether these could lead to the release of Pu into the environment, is therefore a critical part of the safety case for a GDF.

In the UK, Pu is currently stored as PuO₂ powders [4]. These will eventually be processed into a suitable long-term disposal form, for which the two current candidates are disposal MOX [5], where Pu is incorporated into a mixed oxide with U or Th, and zirconolite [6]. However, this is likely to take significant time, during which the PuO₂ may be exposed to a variety of environmental conditions in

interim storage. A major concern is the radiolytic production of H_2 and O_2 from H_2O [7–9]. Higher water content in the powders leads to increased gas production [10], with an almost linear rise in the rates of H_2 and O_2 generation with the number of monolayers above the threshold coverage [11]. Even without external ingress, it is possible for PuO_2 to undergo self-radiolysis [12], whereby small amounts of H_2O introduced during packaging form H_2 , O_2 and H_2O . Depending on the relative humidity during packaging, this can increase the O_2 concentration from 1% to the lower limit of flammability in less than 25 days [11]. PuO_2 is also susceptible to corrosion *via* the oxidation reaction $\text{PuO}_2 + x \text{H}_2\text{O} \rightarrow \text{PuO}_{2+x} + x \text{H}_2$, which also produces H_2 [13, 14]. Gas production and pressurisation of storage canisters, and/or potential exothermic reactions with radiolysis products, could result in premature failure of canisters and other containment mechanisms.

A characteristic feature of PuO_2 powders is a large surface area [4], and the reactivity is determined by the exposed facets and surface speciation. For radiolytic gas production, the key parameter is the adsorption of H_2O . H_2O adsorbs strongly to the surfaces below 120 °C, and progressively desorbs up to 200 °C [15]. H_2O mainly adsorbs through a dissociative mechanism, but can also be present as molecular H_2O [16]. Theoretical studies have shown that dissociative adsorption is favoured at the {100} and {110} surfaces while molecular adsorption is favoured at the {111} [17, 18]. The formation of reactive radicals such as $\text{H}^\bullet$ and $\text{OH}^\bullet$ has also been found to enhance the surface activity of PuO_2 [19, 20]. Understanding the influence of environmental conditions on the morphology and reactivity of PuO_2 powders is therefore key to optimising interim storage conditions to limit undesirable chemical processes.

Most studies have focused on water adsorption in isolation and have not accounted for the potential impact of interactions with other species. For example, Cl^- , which is produced by the degradation of PVC packaging, binds strongly to surfaces [21] and interacts synergistically with H_2O [22], potentially resulting in enhanced H_2O adsorption and radiolytic gas production. Similarly, water ingress due to the failure of storage canisters would expose stored PuO_2 to dissolved CO_2 . We have previously shown that CO_2 can interact strongly with PuO_2 *via* a carbonate species, and can significantly influence the equilibrium particle morphology [23]. There have been conflicting reports that CO_2 can enhance [24–27] or reduce the rate of environmental dispersal of UO_2 [28, 29], but its impact on PuO_2 is unknown [23, 30]. We recently considered the co-absorption of H_2O and CO_2 and found that the presence of both species can stabilise different particle morphologies to those formed in the presence of single species [30].

Understanding the influence of environmental conditions on the particle morphology, surface speciation and reactivity is of critical importance to the nuclear industry, and to other fields more generally, but this topic is largely unexplored for actinides due to the inherent challenges posed by their radioactivity and toxicity [31–34]. Indeed, the issues outlined above extend across the entire nuclear fuel cycle, from fuel synthesis, to burn up, where materials may be exposed to highly non-equilibrium conditions, and to end-of-life disposal.

In this study, we use density-functional theory (DFT) calculations to investigate the interactions of H_2O , CO_2 and H_2O_2 with the stoichiometric and oxygen-deficient {100}, {110} and {111} surfaces of PuO_2 . We then utilise a thermodynamic model developed in previous work to explore the impact of temperature, oxygen stoichiometry and the partial pressures of the three adsorbates on the particle morphology and surface speciation, providing valuable insight towards identifying optimal conditions for safe long-term storage.

2. Computational methodology

Bulk PuO_2 and Pu_2O_3 were modelled as in our previous work [30]. Surface models were generated using the METADISE code [35] and the slab approach. The Miller index of choice was reoriented along the Cartesian z direction, so that the surface lies in the xy plane, and a 15 Å vacuum gap was inserted between periodic images along z to form identical top and bottom surfaces. Models of the {100}, {110} and {111} surfaces, corresponding respectively to $\sqrt{2} \times \sqrt{2}$, $2 \times \sqrt{2}$ and $\sqrt{2} \times \sqrt{2}$ expansions of the surface repeat units, were built with either seven layers ({100}/{110}, 28 PuO_2 units) or five layers ({111}, 20 units). In the {100} surface model, half the oxygen atoms were transposed from the top to the bottom surface to quench the dipole. To create oxygen-deficient surface models, oxygen vacancies were introduced at both the top and bottom surfaces. Each oxygen vacancy introduces two excess electrons, which localise to surface Pu to create two Pu^{3+} . We do not constrain the position of the Pu^{3+} , but our setup ensures that the inversion symmetry of the slab is preserved.

Table 1. List of surface adsorption configurations examined in this work with the scheme used to label them. $\{hkl\} = \{100\}$, $\{110\}$ or $\{111\}$ specifies the Miller indices of the surface and X = S or R specifies a stoichiometric (S) and O-deficient (reduced, R) surface. The adsorbed species are followed by single-letter codes indicating associative (A), dissociative (D) or molecular (M) adsorption. For example, the dissociative adsorption of H₂O on the stoichiometric $\{110\}$ is denoted $\{110\}$ S-H₂O-D, and the co-adsorption of dissociatively-adsorbed H₂O and molecularly-adsorbed H₂O₂ on the O-deficient $\{111\}$ is denoted $\{111\}$ R-H₂O-H₂O₂-DM. Note that data for some of the models is taken from our previous studies [23, 30].

Surface	Adsorption description
Bare surface	$\{hkl\}$ X
Single adsorption of CO ₂ [23]	$\{hkl\}$ X-CO ₂ -A
Single adsorption of H ₂ O [30]	$\{hkl\}$ X-CO ₂ -M
	$\{hkl\}$ X-H ₂ O-D
Single adsorption of H ₂ O ₂	$\{hkl\}$ X-H ₂ O-M
	$\{hkl\}$ X-H ₂ O ₂ -D
Co-adsorption of H ₂ O and CO ₂ [30]	$\{hkl\}$ X-H ₂ O ₂ -M
	$\{hkl\}$ X-H ₂ O-CO ₂ -DA
	$\{hkl\}$ X-H ₂ O-CO ₂ -DM
	$\{hkl\}$ X-H ₂ O-CO ₂ -MA
	$\{hkl\}$ X-H ₂ O-CO ₂ -MM
Co-adsorption of H ₂ O ₂ and CO ₂	$\{hkl\}$ X-H ₂ O ₂ -CO ₂ -DA
	$\{hkl\}$ X-H ₂ O ₂ -CO ₂ -DM
	$\{hkl\}$ X-H ₂ O ₂ -CO ₂ -MA
	$\{hkl\}$ X-H ₂ O ₂ -CO ₂ -MM
Co-adsorption of H ₂ O and H ₂ O ₂	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -DD
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -DM
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -MD
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -MM
Co-adsorption of H ₂ O, H ₂ O ₂ and CO ₂	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -CO ₂ -DDA
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -CO ₂ -DDM
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -CO ₂ -DMA
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -CO ₂ -DMM
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -CO ₂ -MDA
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -CO ₂ -MDM
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -CO ₂ -MMA
	$\{hkl\}$ X-H ₂ O-H ₂ O ₂ -CO ₂ -MMM

Reference calculations were performed on isolated molecules of H₂O₂, CO₂, H₂O and O₂ by placing them at the centre of a 10 Å³ cell. The vacuum gap in the slab models, and the cell size for the calculations on molecules, were chosen to avoid interactions between periodic images.

Surface models with (co-)adsorbed H₂O, H₂O₂ and CO₂ were built with a range of different starting configurations to sample the structure and energetics of adsorption. Due to the complex adsorption landscape, we only considered configurations that maximise the hydrogen-bonding network and surface-adsorbate interactions, and which result in the most favourable (i.e. most negative) adsorption energies. Table 1 lists the models examined in this work and explains the labelling scheme used throughout the text. Data for the single and co-adsorption of H₂O and CO₂ are taken from our previous work [23, 30].

The configurations include associative (A), dissociative (D), and molecular (M) adsorption. We consider molecular adsorption for all three species, where the molecule binds to the surface intact. For H₂O and H₂O₂, we also consider dissociative adsorption as H/OH and HO/OH respectively. While H₂O₂ can potentially dissociate as HO/OH, H/OOH, and H/OO/H, we focussed on the first possibility since OH groups are common to dissociatively-adsorbed H₂O. The H atoms from dissociatively-adsorbed H₂O typically adsorb to surface O, while the hydroxy groups from H₂O and H₂O₂ adsorb to surface Pu or ‘heal’ oxygen vacancies at O-deficient surfaces. For CO₂, we consider associative rather than dissociative adsorption, where the CO₂ molecule binds to a surface O atom to form a carbonate species. The surface coverage in our models ranges from 0.25 to 0.75 from the single to the three-species adsorption configurations, respectively, in increments of 0.25, each of which represents the addition of one adsorbate per surface Pu atom. This fixed coverage was chosen to keep the configurational landscapes for the adsorption manageable. We highlight considering different levels of coverage, and also other potential adsorbates such as O₂ and NO [36], as potential targets for future work.

DFT calculations were performed using the Vienna *Ab initio* simulation package [37]. The ion cores were modelled using projected augmented-wave pseudopotentials [38, 39] with frozen cores of [He] and [Rn] for O/C and Pu, respectively. Spin-orbit interactions (SOI) were considered in all calculations, and the magnetic structure was based on the 3k non-collinear antiferromagnetic ordering of PuO₂ [23, 30, 40]. A plane-wave basis set with a 500 eV cutoff energy was used to model the valence electronic structure. The Perdew–Burke–Ernzerhof generalised-gradient approximation exchange–correlation functional was used with a Hubbard U correction applied to the Pu 5f states using the method of Liechtenstein *et al* [41]. In this approach, the Coulomb and exchange parameters U and J are treated as independent variables. We set $U = 6$ eV and $J = 0$, which makes the parameterisation equivalent to the Dudarev method [42], to match previous literature [23, 30, 40]. We note that we have not included a dispersion correction in our study. There have been no comprehensive studies of the effects of vdW corrections on the adsorption of small molecules to actinide oxide surfaces, nor on the interaction of various vdW correction schemes with SOI and non-collinear magnetism. The fact that there are multiple implementations of dispersion corrections further complicates this task [43]. However, we note that one of the few studies that have tested the DFT-D3 dispersion correction [44] for the adsorption of H₂O to PuO₂ surfaces, albeit without SOI, found a consistent decrease in the adsorption energies that did not change the stability order of the configurations tested [18]. We therefore highlight comprehensive testing of dispersion corrections as another avenue for future development.

Γ -centred Monkhorst–Pack $\mathbf{k}$ -point grids with $5 \times 5 \times 5$, $4 \times 4 \times 4$ and $2 \times 2 \times 1$ subdivisions were used to sample the electronic Brillouin zones of bulk PuO₂, Pu₂O₃ and the slab models, respectively, while a mesh with $\mathbf{k} = \Gamma$ was used for the calculations on isolated molecules. All geometry optimisations were performed using the conjugate gradient algorithm with thresholds of 10^{-5} eV and 10^{-2} eV Å⁻¹ on the electronic total energy and ionic forces.

Analyses were performed using the SurfinPy [45, 46] and Pymatgen Python packages [47].

3. Results and discussion

3.1. Structure and energetics H₂O₂ adsorption and co-adsorption with H₂O and CO₂

Each of the stoichiometric and O-deficient surface models has exposed O and Pu atoms to which adsorbates can interact either by formation of a direct bond or through e.g. hydrogen bonding. The {100}, {110} and {111} surfaces differ in the position of the Pu atoms relative to the surface O, with the Pu being most recessed at the {111} and most accessible at the {110}. The O-deficient surfaces have both Pu³⁺ and Pu⁴⁺ sites, as well as surface oxygen vacancies (V_O) that can be healed by adsorbates, and therefore support more flexible adsorption than the stoichiometric surfaces. The presence of multiple binding sites, and the possibility of both molecular and dissociative/associative adsorption, leads to a relatively complex adsorption landscape. The possibility for interactions between multiple adsorbates adds further complexity to the adsorption landscapes of co-adsorption configurations. Across the three stoichiometric and reduced surfaces, and the different combinations of up to three species, we examined over 100 adsorption configurations. In view of this, we focus our discussion on more general trends, but we provide data for all our configurations in the Supporting Information and our optimised structures in a data repository [48] (see data availability statement at the end of the manuscript).

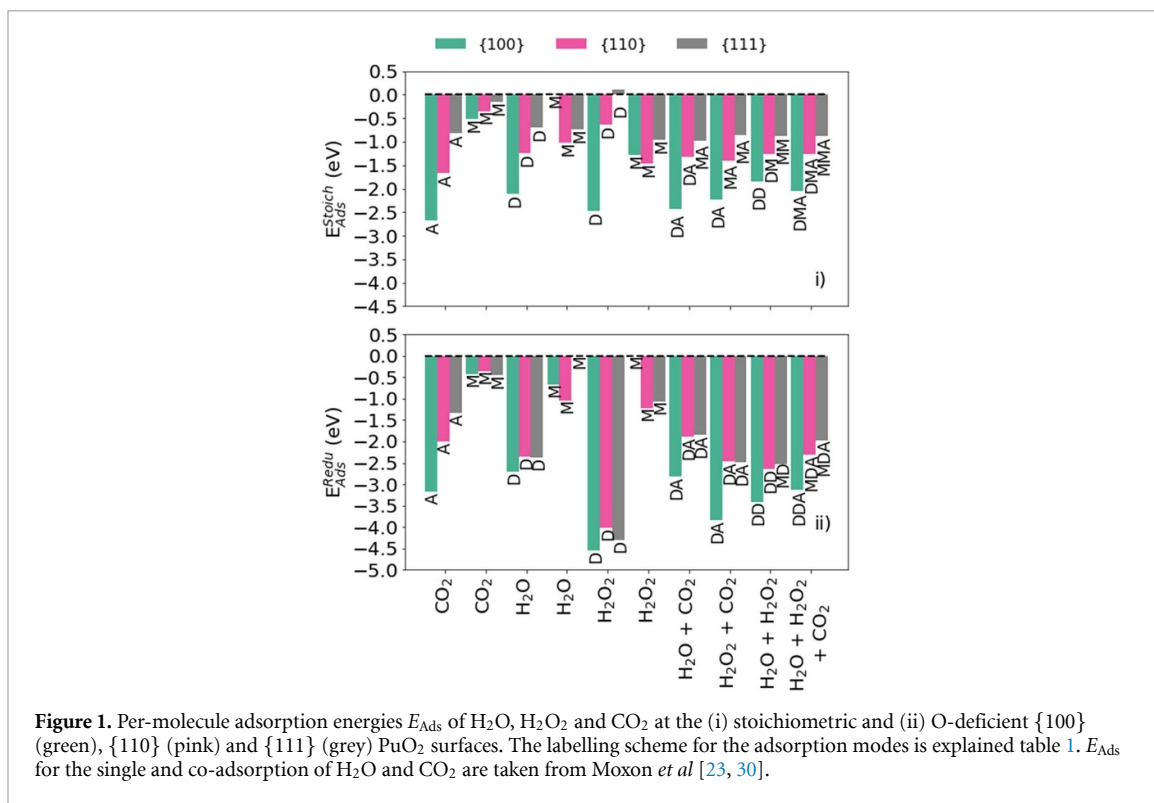
The strength of the interaction between adsorbed molecules and a surface is defined by the adsorption energy, E_{Ads} , given by:

$$E_{\text{Ads}} = \frac{E_{\text{Slab,Ads}} - (E_{\text{Slab,Bare}} + \sum_i n_i E_i)}{\sum_i n_i} \quad (1)$$

where $E_{\text{Slab,Ads}}$ and $E_{\text{Slab,Bare}}$ are the energies of the slab model with n_i species i adsorbed to the surface and the reference bare surface, respectively, and E_i are the energies of the isolated molecule(s). A more negative E_{Ads} implies stronger binding.

The E_{Ads} of the stable configurations listed in table 1 are summarised in figure 1. Data for the single and co-adsorption of H₂O and CO₂ are taken from our previous work [23, 30], and for brevity, we only consider the configurations with the lowest E_{Ads} .

Formation of a direct bond to the surface through associative or dissociative adsorption almost invariably results in more negative E_{Ads} than molecular adsorption. The sole exceptions are the adsorption of H₂O at the stoichiometric {111} and the adsorption of H₂O₂ at the stoichiometric {110} and {111}. In particular, the dissociative adsorption of H₂O₂ at the stoichiometric {111} has a positive adsorption energy (i.e. is energetically unfavourable) due to the formation of a pair of OH• radicals. This has also been observed for CeO₂ surfaces [49, 50]. The E_{Ads} generally follow the trend



{100} < {110} < {111} on both the stoichiometric and O-deficient surfaces, i.e. adsorption at the {100} is energetically preferred. There are a handful of exceptions to the latter, *viz* that CO_2 shows comparable E_{Ads} for molecular adsorption at all three stoichiometric surfaces, and molecular adsorption of H_2O and H_2O_2 is most favourable at the stoichiometric and O-deficient {110}. However, in all cases associative and dissociative absorption are energetically preferred and do follow the general trend. We also find that adsorption is almost always more favourable at the O-deficient surfaces, indicating that the presence of surface oxygen vacancies (V_{O}) promotes stronger binding. The only notable exception is again that some of the molecular adsorption configurations are marginally more favourable at the stoichiometric surfaces, although in all cases we predict that associative/dissociative binding would be energetically favoured.

As we have examined the single- and co-adsorption of H_2O and CO_2 in detail in previous work, we focus here on the configurations with H_2O_2 . While there have been DFT studies of the adsorption of H_2O and H_2 at PuO_2 surfaces, to our knowledge there are no studies of H_2O_2 and the behaviour of this species is thus largely unknown [10]. However, H_2O_2 has been shown to increase the dissolution rate of PuO_2 [51], and so understanding how H_2O_2 interacts with PuO_2 surfaces is crucial. Our calculations predict that H_2O_2 shows distinct dissociative and molecular adsorption modes (figure 2). Dissociative adsorption is energetically preferred at the stoichiometric {100} and at all three O-deficient surfaces, whereas the stoichiometric {110} and {111} favour molecular absorption. For dissociative adsorption at the stoichiometric surfaces, the two OH bond to different surface Pu atoms, whereas on the O-deficient surfaces one of the OH ‘heals’ the V_{O} and the other bonds to a surface Pu as for the stoichiometric surfaces.

The propensity of adsorbates to heal surface V_{O} in this way has been widely reported in the literature [30, 49, 50, 53, 54]. The behaviour of H_2O_2 predicted by our calculations is comparable to studies of the molecular adsorption of H_2O_2 on UO_2 [55], which reported similar binding geometries and E_{Ads} and a similar trend of {100} $\approx$ {110} < {111} (i.e. similar and more negative E_{Ads} at the {100} and {110} compared to the {111}).

Under typical environmental conditions, H_2O_2 is unlikely to be adsorbed in isolation. Figure 3 compares the most favourable geometries in the pairwise and three-species co-adsorption configurations, i.e. H_2O_2 with CO_2 or H_2O , and H_2O_2 with CO_2 and H_2O , at the stoichiometric and O-deficient {100}, and the full set of optimised geometries for the two- and three-species configurations are shown in figures S1-S3. Broadly, the binding geometries in the co-adsorption configurations are similar to those adopted by the molecules in the individual adsorption configurations, with the species arranged to maximise the hydrogen bonding network and the interactions with the surface, including OH generated

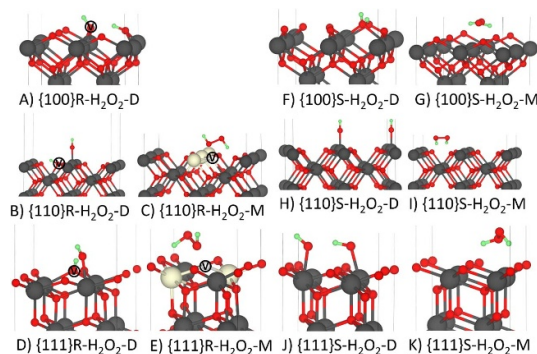


Figure 2. Adsorption geometries of H_2O_2 on PuO_2 surfaces. Each configuration is labelled according to the scheme in table 1. The Pu, O and H atoms are shown in grey, red and green respectively. For the O-deficient surfaces (A–E), the cream atoms indicate the positions of the Pu^{3+} and the circles marked 'V' show the oxygen vacancies. Where the latter overlap with a red sphere this indicates that the vacancy has been 'healed' by an OH from dissociatively-adsorbed H_2O_2 . Created using the VESTA software package [52].

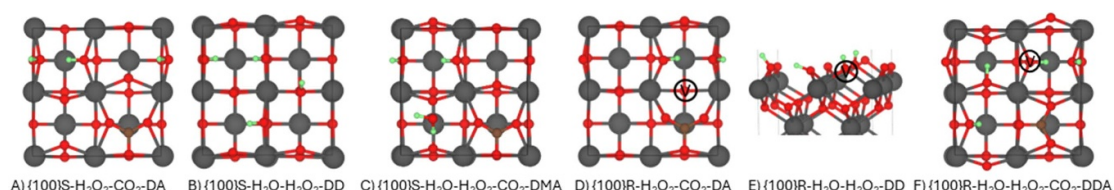


Figure 3. Selected adsorption configurations of co-adsorbed H_2O_2 and CO_2 (A/D), H_2O_2 and H_2O (B/E), and H_2O , H_2O_2 and CO_2 (C/F) at the stoichiometric (A–C) and O-deficient {100} surface of PuO_2 (D–F). Each configuration is labelled according to the scheme in table 1. The Pu, O and H atoms are shown in grey, red and green respectively. For the O-deficient surfaces (D–E), the cream atoms indicate the positions of the Pu^{3+} and the circles marked 'V' show the oxygen vacancies. Where the latter overlap with a red sphere this indicates that the vacancy has been 'healed' by one of the adsorbates. Created using the VESTA software package [52].

by dissociative adsorption of H_2O or H_2O_2 healing the V_O at the O-deficient surfaces. We note here that while we attempted to minimise all possible combinations of dissociative/associative and molecular adsorption of the adsorbates, several combinations were unstable and spontaneously minimised to alternative configurations.

The E_{Ads} for co-adsorption follow a similar trend to individual species, with a typical ordering of $\{100\} < \{110\} < \{111\}$, more favourable binding to the O-deficient surfaces, and a general preference for associative/dissociative rather than molecular adsorption. However, some species are predicted to show molecular adsorption in some co-adsorption configurations, *viz.* H_2O and H_2O_2 at some of the stoichiometric surfaces, and H_2O at some of the O-deficient surfaces.

The adsorption energies in figure 1 are per adsorbate, which allows the E_{Ads} of the singly- and co-adsorbed species to be compared. A more negative per-molecule E_{Ads} for a co-adsorption configuration indicates a synergistic effect (*i.e.* the additional species favours the adsorption), while a more positive E_{Ads} indicates an antagonistic effect (*i.e.* the additional species hinders the co-adsorption). Co-adsorption with H_2O_2 at the stoichiometric {100} and {110} has an antagonistic effect on the binding of CO_2 but a synergistic effect at the {111}. In all three cases, the co-adsorption raises the per-molecule E_{Ads} compared to the single adsorption of H_2O_2 , although the effect is relatively small, particularly at the {110} and {111}. On the other hand, co-adsorption at all three O-deficient surfaces favours the adsorption of CO_2 but significantly increases the per-molecule E_{Ads} compared to singly-adsorbed H_2O_2 . Our calculations predict that co-adsorption of H_2O and H_2O_2 leads to higher per-molecule E_{Ads} than either species in isolation at the stoichiometric {100}, but favours the adsorption of H_2O_2 at the {110} and H_2O at the {111} with minimal impact on the other co-adsorbate. The effect of co-adsorption at the O-deficient surfaces is more pronounced and favours/disfavours the adsorption of $\text{H}_2\text{O}/\text{H}_2\text{O}_2$ relative to single adsorption.

Compared to the single species, co-adsorption of all three species raises the per-molecule E_{Ads} of CO_2 and H_2O_2 at the stoichiometric {100} and {110}, but has synergistic and antagonistic effects on the binding of CO_2 and H_2O , respectively, at the stoichiometric {111}. At the O-deficient surfaces, the

co-adsorption of all three species consistently disfavours the adsorption of H_2O_2 , selectively favours/disfavours the adsorption of H_2O at the $\{100\}/\{111\}$ and disfavours the adsorption of CO_2 at the $\{110\}$ and $\{111\}$. Compared to the two-species configurations, the addition of H_2O_2 increases the E_{Ads} of H_2O and CO_2 at the stoichiometric $\{100\}$, the addition of H_2O disfavours the adsorption of CO_2 and H_2O_2 at the stoichiometric $\{100\}$ and $\{110\}$, and the addition of CO_2 decreases the E_{Ads} of H_2O and H_2O_2 at the stoichiometric $\{100\}$. H_2O_2 favours the co-adsorption H_2O and CO_2 at all three O-deficient surfaces, whereas H_2O increases the E_{Ads} of co-adsorbed CO_2 and H_2O_2 , and CO_2 increases the E_{Ads} of co-adsorbed H_2O and H_2O_2 . Notably, whereas dissociative adsorption of singly-adsorbed H_2O is strongly preferred at all three O-deficient surfaces, H_2O adopts a molecular binding geometry in the lowest-energy co-adsorption configuration with H_2O_2 at the O-deficient $\{111\}$ and when co-adsorbed with H_2O_2 and CO_2 at the O-deficient $\{110\}$ and $\{111\}$. The dissociative adsorption of H_2O_2 has a more negative E_{Ads} than the equivalent adsorption of H_2O at all three O-deficient surfaces, which, together with the observation that H_2O adopts molecular geometries in some of the co-adsorption configurations with H_2O_2 , suggests the latter has a stronger propensity for healing surface V_{O} .

3.2. Desorption temperatures

We first consider the effect of temperature on the desorption of CO_2 , H_2O and H_2O_2 from the stoichiometric and O-deficient $\{100\}$, $\{110\}$ and $\{111\}$ surfaces. The desorption temperature $T_{\text{D},i}$ defines the highest temperature required to desorb an adsorbate i from a surface at a given adsorbate partial pressure p_i , with higher T_{D} for a given p_i indicative of stronger adsorption.

To predict the T_{D} we use the methodology employed successfully in previous studies [17, 23, 56–59]. The surface energies γ_{Bare} are calculated as:

$$\gamma_{\text{Bare}} = \frac{1}{2A} \{E_{\text{Slab,Bare}} - [(n_{\text{PuO}_2} - 2n_{\text{V}})E_{\text{Bulk,PuO}_2} + n_{\text{V}}E_{\text{Bulk,Pu}_2\text{O}_3}]\} \quad (2)$$

where $E_{\text{Slab,Bare}}$ is the energy of the bare stoichiometric or O-deficient slabs, composed of n_{PuO_2} PuO_2 formula units and n_{V} oxygen vacancies, $E_{\text{Bulk,PuO}_2}$ and $E_{\text{Bulk,Pu}_2\text{O}_3}$ are the energies per formula unit of bulk PuO_2 and Pu_2O_3 , and A is the surface area of one of the two identical surfaces of the slab model (and thus $2A$ is the total surface area). For surfaces with an adsorbate of interest i , the corresponding surface energies $\gamma_{\text{Ads},i}$ are:

$$\gamma_{\text{Ads},i} = \gamma_{\text{Bare}} + \Gamma_i \left[E_{\text{Ads},i}(T) - RT \ln \left(\frac{p_i}{p^\circ} \right) \right] \quad (3)$$

$\Gamma_i = n_i/2A$ is the surface coverage of the adsorbate i . The temperature-dependent adsorption energy $E_{\text{Ads},i}$ is obtained by modifying equation (1) to:

$$E_{\text{Ads},i}(T) = \frac{E_{\text{Slab,Ads}} - [E_{\text{Slab,Des}} + n_i E_i(T)]}{n_i} \quad (4)$$

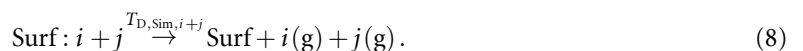
$E_{\text{Slab,Des}}$ is the energy of the slab with molecule i desorbed, which may be a bare surface for a single adsorbate ($E_{\text{Slab,Des}} = E_{\text{Slab,Bare}}$) or a surface with one or more co-adsorbate(s) remaining. We also use temperature-dependent adsorbate energies $E_i(T)$ including an entropy correction $TS_i(T)$ for the gas-phase species using data from the NIST database [60]:

$$E_i(T) = E_i - TS_i(T). \quad (5)$$

For surfaces with a single adsorbate i , the T_{D} correspond to the desorption process given by:



For surfaces with two adsorbates i and j , we can define two different T_{D} : a $T_{\text{D, Sel}, i}$ for selective desorption of one species i while the other species j remains adsorbed, and a $T_{\text{D, Sim}, i+j}$ for simultaneous desorption of both species. These two T_{D} correspond to the desorption processes given by:



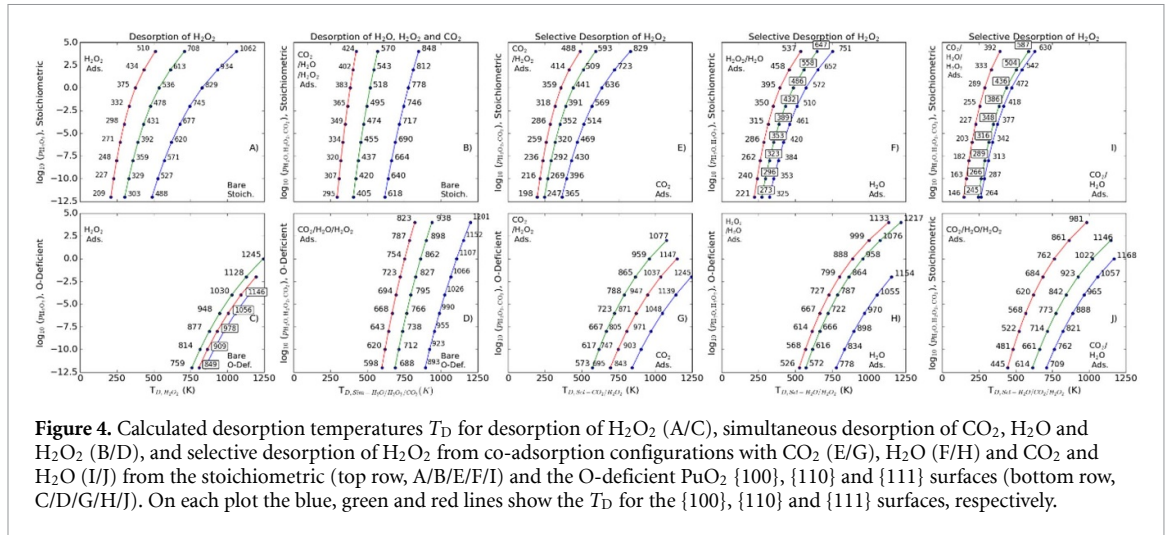


Figure 4. Calculated desorption temperatures T_D for desorption of H_2O_2 (A/C), simultaneous desorption of CO_2 , H_2O and H_2O_2 (B/D), and selective desorption of H_2O_2 from co-adsorption configurations with CO_2 (E/G), H_2O (F/H) and CO_2 and H_2O (I/J) from the stoichiometric (top row, A/B/E/F/I) and the O-deficient PuO_2 {100}, {110} and {111} surfaces (bottom row, C/D/G/H/J). On each plot the blue, green and red lines show the T_D for the {100}, {110} and {111} surfaces, respectively.

To calculate the $T_{D,\text{Sel}}$ in equation (7), we consider only the coverage Γ_i and partial pressure p_i of the desorbing species. To calculate the $T_{D,\text{Sim}}$ in equation (8), we calculate the total adsorbate coverage $\Gamma = (\sum_i n_i)/2A$ and partial pressure $p_{\text{Ads}} = \sum_i p_i$. These formulae are easily generalised to the configurations with three (or more) co-adsorbates.

The T_D are best visualised on a two-dimensional plot of temperature against partial pressure of the desorbing species, on which the T_D delineates a phase boundary between surfaces with and without the adsorbate i . As in the previous section, we focus here on the behaviour of H_2O_2 and its co-adsorption with other species. Selected T_D predictions are shown in figure 4, and the full set of calculated T_D are provided in figure S4.

Figure 4(A/C) shows the T_D of H_2O_2 from the stoichiometric and O-deficient surfaces. For the stoichiometric surfaces, we predict $T_{D,\text{H}_2\text{O}_2}$ in the order of {100} > {110} > {111}, which reflects the ordering of the adsorption energies in figure 1. We predict higher, but also more similar $T_{D,\text{H}_2\text{O}_2}$ for the O-deficient surfaces, reflecting the similar and larger E_{Ads} compared to those calculated for the stoichiometric surfaces, with an ordering of {100} > {111} > {110}.

Figures 4(B/D) and S4 show predictions for the $T_{D,\text{Sim}}$ for simultaneous desorption of pairs of adsorbates from the three two-species co-adsorption configurations, including data from our previous study on the co-adsorption of CO_2 and H_2O [30], and for the simultaneous desorption of all three adsorbates from the three-species configuration. The $T_{D,\text{Sim}}$ of multiple species from the stoichiometric surfaces invariably follow the order {100} > {110} > {111}. For the O-deficient surfaces, the $T_{D,\text{Sim}}$ are always predicted to be highest at the {100}, in most cases by a wide margin, with similar $T_{D,\text{Sim}}$ at the {110} and {111}. The $T_{D,\text{Sim}}$ for all three O-deficient surfaces are also higher than the $T_{D,\text{Sim}}$ predicted for any of the three stoichiometric surfaces. We note that these findings again reflect the predicted per-molecule E_{Ads} in figure 1.

The $T_{D,\text{Sel}}$ for the selective desorption of a molecule when co-adsorbed with other species provides an alternative interpretation of how the co-adsorption affects its binding. It is useful to define a change in desorption temperature $\Delta T_{D,\text{Sel}}$ which for a two-species configuration is given by:

$$\Delta T_{D,\text{Sel},i,j} = T_{D,\text{Sel},i,j} - T_{D,i} \quad (9)$$

where i and j denote the desorbing and remaining molecule respectively. $\Delta T_D > 0$ and $\Delta T_D < 0$ indicate that the co-adsorbate(s) stabilise or destabilise the binding of species i , while a $\Delta T_D \approx 0$ indicates that the additional species have a minimal effect.

The predicted $T_{D,\text{Sel}}$ for the selective desorption of H_2O_2 when co-adsorbed with CO_2 and/or H_2O are shown in figure 4(E-J), and a full set of predicted ΔT_D are provided in figure S5. At the stoichiometric {100} and {110}, we predict that co-adsorption of H_2O_2 with either or both co-adsorbates lowers the $T_{D,\text{Sel}}$ of all the adsorbates, including H_2O_2 . At a reference $p_i = 1$ bar of the desorbing molecule, we predict ΔT_D of up to -356 and -141 K at the two surfaces. These extremes correspond to selective desorption of H_2O_2 and CO_2 , respectively, from $\text{H}_2\text{O}_2/\text{H}_2\text{O}/\text{CO}_2$. The calculated T_D and $T_{D,\text{Sel}}$ in the first case are 829 and 472 K, and the presence of co-adsorbed H_2O and CO_2 therefore significantly influences the desorption of H_2O_2 . The antagonistic effects of the co-adsorption are more pronounced at higher partial pressures of the desorbing molecules, which, while seemingly counter-intuitive, can be understood with

reference to the calculated T_D and $T_{D,sel}$ (cf figures S4 and S5): higher p_i increase both desorption temperatures, as expected, but have a larger impact on the T_D for single desorption, which leads to a more negative ΔT_D . In contrast to the {100} and {110}, co-adsorption is predicted to have a small (positive or negative) impact on desorption from the stoichiometric {111}, with $\Delta T_D < 50$ K except for the selective desorption of H_2O and H_2O_2 from the three-species configurations ($\Delta T_D \approx 100$ K at high p_{H_2O} and $\Delta T_D \approx -120$ K at high $p_{H_2O_2}$ respectively). Co-adsorption therefore has a lower impact on the desorption of species from the stoichiometric {111}, likely because of the weaker binding (lower E_{Ads}) at this surface compared to the {100} and {110}.

The $T_{D,sel}$ and ΔT_D at the O-deficient surfaces are typically higher than at the stoichiometric surfaces. Co-adsorption at the O-deficient surfaces invariably results in negative ΔT_D , meaning that adsorbates are more easily desorbed than when present as a single species, with the largest impacts at the {111} and {110}.

At $p_i = 1$ bar, we predict large negative ΔT_D of up to -614 and -454 K for the selective desorption of H_2O from H_2O_2/H_2O at the O-deficient {111} and {110}. The corresponding (lowered) $T_{D,sel}$ are 370 and 574 K, indicating that co-adsorption of H_2O_2 significantly destabilises the binding of H_2O at these surfaces. The ΔT_D of H_2O_2 are large but lower at -430 and -286 K. With reference to figure 1, the E_{Ads} for singly-adsorbed H_2O_2 is larger than the E_{Ads} for H_2O and the per-molecule E_{Ads} for co-adsorbed H_2O_2/H_2O , so desorbing H_2O results in a larger energetic gain and, consequently, a larger negative ΔT_D . The effects of co-adsorption on the T_D for desorption from the O-deficient {100} are, on the other hand, generally less pronounced, and the ΔT_D are in many cases lower than for the equivalent stoichiometric surfaces. The two exceptions in our calculations are the ΔT_D for desorption of H_2O and CO_2 from $CO_2/H_2O/H_2O_2$, for which the $T_{D,sel}$ for the stoichiometric and O-deficient {100} are similar but the ΔT_D for the O-deficient surface are $2-3\times$ larger. This again reflects the generally stronger adsorption at the O-deficient surfaces and the larger impact of co-adsorption on the E_{Ads} .

The co-adsorption of three species tends to have a smaller impact on the T_D for desorption from the stoichiometric {100}, but comparable or larger impacts at the stoichiometric {110} and {111}. The one exception to this trend in our calculations is that the presence of both H_2O and CO_2 has a larger destabilising effect on H_2O_2 at the {100} than either of the two-species configurations. For the O-deficient surfaces, this trend tends to be reversed, such that the effect on the T_D for desorption from the {111} and {110}, for which the ΔT_D are largest, tends to be lower in the three-species configurations, whereas the effect on the {100}, where the ΔT_D are smallest, tends to be enhanced. The exception again is the desorption of H_2O_2 from $CO_2/H_2O/H_2O_2$ at the O-deficient {111}, for which the ΔT_D are larger than those for desorption of H_2O_2 from CO_2/H_2O_2 and H_2O_2/H_2O at the same surface.

Finally, it is interesting to compare the $T_{D,sim}$ for simultaneous desorption of two or three co-adsorbates to the $T_{D,sel}$ for the selective desorption of individual species. If the $T_{D,sim}$ are lower than all the $T_{D,sel}$, it is more favourable for the adsorbates to desorb together and leave a bare surface. If this is not the case, then the species with the lower $T_{D,sel}$ will selectively desorb first. Table 2 provides a summary of the T_D for desorption from the stoichiometric and O-deficient surfaces calculated at a reference $p_i = 1$ bar of the desorbing molecule(s). In almost all cases we predict that the $T_{D,sim}$ are higher than one or more $T_{D,sel}$, which means that one species will selectively desorb first as the temperature is raised. The one exception is the stoichiometric {111} with co-adsorbed H_2O and H_2O_2 , where the $T_{D,sim}$ are a very small 2 and 12 K lower, respectively, than the $T_{D,sel}$ of the individual species.

When H_2O_2 is co-adsorbed with CO_2 or CO_2 and H_2O at the stoichiometric surfaces, we predict that H_2O_2 will selectively desorb first. When H_2O_2 is co-adsorbed with just H_2O , on the other hand, we predict that H_2O will selectively desorb from the {100} and simultaneous desorption will occur at the {110} and {111}. When H_2O_2 is co-adsorbed with either H_2O or CO_2 at the three O-deficient surfaces, we predict that the co-adsorbate will selectively desorb first. When all three species are co-adsorbed, we predict that CO_2 will desorb first from the O-deficient {110} and {111} surfaces, while H_2O will desorb first from the {100}. The predicted T_D for the resultant two-species configurations further indicate that H_2O_2 will be the last molecule to desorb, which once again highlights the strong affinity of the O-deficient surfaces for H_2O_2 due to its higher propensity to heal surface oxygen vacancies.

3.3. Surface composition

Using equation (2), we calculate surface energies of $\gamma_{Bare} = 1.98, 1.36$ and 0.95 J m $^{-2}$ for the bare stoichiometric {100}, {110} and {111}, respectively, and $1.90, 1.38$ and 1.22 J m $^{-2}$ for the equivalent O-deficient surfaces [23]. These are in good agreement with previous literature [61]. An important observation is that the surface energies γ_{Bare} fall in the order {111} < {110} < {100} (i.e. the {111} is the most stable bare surface), which largely opposes the trend in the adsorbate binding energies in figure 1 where

Table 2. Desorption temperatures T_D for the desorption of singly-adsorbed H_2O_2 and the simultaneous and selective desorption of the species in co-adsorption configurations with H_2O_2/H_2O , CO_2/H_2O_2 and $CO_2/H_2O/H_2O_2$ at the stoichiometric and O-deficient {100}, {110} and {111} surfaces of PuO_2 , calculated at a reference $p = 1$ bar of the desorbing molecule(s).

	Desorbing molecule(s)	Temperature (K)					
		Stoichiometric surfaces			O-deficient surfaces		
		{100}	{110}	{111}	{100}	{110}	{111}
T_{D,H_2O_2}	H_2O_2	829	536	375	1380	1245	1318
$T_{D,Sim,H_2O_2+H_2O}$	$H_2O_2 + H_2O$	717	522	383	1200	969	933
$T_{D,Sel,H_2O:H_2O_2}$	H_2O	557	495	385	948	574	370
$T_{D,Sel,H_2O_2:H_2O}$	H_2O_2	572	486	395	1274	958	888
$T_{D,Sim,CO_2+H_2O_2}$	$CO_2 + H_2O_2$	798	544	360	1250	865	871
$T_{D,Sel,CO_2:H_2O_2}$	CO_2	757	547	340	1100	390	304
$T_{D,Sel,H_2O_2:CO_2}$	H_2O_2	636	441	359	1371	959	1147
$T_{D,Sim,H_2O+H_2O_2+CO_2}$	$H_2O + H_2O_2 + CO_2$	778	518	383	1107	862	754
$T_{D,Sel,H_2O:CO_2+H_2O_2}$	H_2O	720	447	430	726	845	437
$T_{D,Sel,H_2O_2:CO_2+H_2O}$	H_2O_2	472	436	289	1168	1022	762
$T_{D,Sel,CO_2:H_2O+H_2O_2}$	CO_2	891	506	380	917	638	365

the strongest adsorption energy is generally for species adsorbed on the {100} surface. As shown in previous work [30, 54], this means that the presence of adsorbates can significantly influence the equilibrium surface expression due to the E_{Ads} from adsorbate binding offsetting the differences in γ_{Bare} , following equation (3).

The surface free energies can be used to predict the equilibrium surface composition as a function of temperature and adsorbate partial pressure. We first generalise equation (3) for multiple adsorbates:

$$\gamma_{Ads}(T, \mu_i) = \frac{1}{2A} \{E_{Slab} - [(n_{PuO_2} - 2n_V)E_{Bulk, PuO_2} + n_VE_{Bulk, Pu_2O_3}]\} - \sum_i \Gamma_i [\mu_i^\circ(T) + \Delta\mu_i] \quad (10)$$

where E_{Slab} is the energy of the slab model, and μ_i° and $\Delta\mu_i$ are respectively the standard chemical potentials of the adsorbates, which we equate to the $E_i(T)$ defined in equation (5), and externally-imposed chemical potential shifts. Under the assumption that the adsorbates behave as ideal gases, the partial pressures p_i can be expressed in terms of the chemical potentials as:

$$p_i = \frac{\mu_i^\circ(T) + \Delta\mu_i}{k_B T} \quad (11)$$

where k_B is the Boltzmann constant. Equation (10) thus allows the impact of varying the temperature and p_i on the surface energies, and hence the equilibrium surface expression (i.e. the area ratios of the {100}, {110} and {111}) to be predicted.

The predictions can be conveniently visualised as surface area ‘phase diagrams’ that display the relative areas of the three surfaces as a function of two variables (T and/or p_i). Figure 5(A/B) shows the predicted surface composition of stoichiometric and O-deficient PuO_2 particles as a function of temperature and $p_{H_2O_2}$. We also superimpose the calculated T_D for the three surfaces to allow the speciation (i.e. bare or with adsorbed H_2O_2) to be visualised. For both stoichiometric and O-deficient particles, we predict dominant expression of the {111} surface, which mirrors the significantly lower γ_{Bare} of this surface. For stoichiometric particles, we predict a small (<1%) expression of the {100} around the T_D of H_2O_2 from the {111}, which highlights a delicate balance of energetics. Below the T_D the favourable surface composition is the most stable {111} surface with adsorbed H_2O_2 . Above the T_D the bare {111} surface is favoured. However, around the T_D it is more favourable for the H_2O_2 desorbed from the {111} to adsorb to the more reactive {100}, which creates a driving force for expressing this surface. For O-deficient particles, the γ_{Bare} of the {110} and {111} are closer in energy, and the E_{Ads} of H_2O_2 are relatively similar (cf figure 1). This results in up to 17.2% expression of the {110} both at low T and high $p_{H_2O_2}$, and over a narrow range of conditions close to the T_D of H_2O_2 , and up to 1.2% expression of the O-deficient {100}.

To explore the effect of the co-adsorption of H_2O_2 with other species, we generate similar diagrams as a function of $p_{H_2O_2}$ and p_{CO_2} or p_{H_2O} , for stoichiometric and O-deficient PuO_2 particles at low and high temperatures of $T = 298$ or 1000 K (figure 6). The presence of H_2O or CO_2 in addition to H_2O_2 results in similar qualitative surface expression, *viz.* dominant {111} with up to 7.5% {100} under some conditions. Co-adsorption of H_2O_2 and CO_2 has a more pronounced effect, with the expression of the

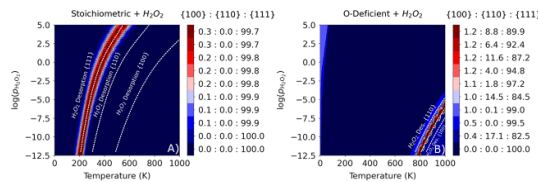


Figure 5. Surface area phase diagrams showing the area ratios of the {100}, {110} and {111} surfaces of stoichiometric (A) and O-deficient PuO_2 particles (B) as a function of temperature T and H_2O_2 partial pressure $p_{\text{H}_2\text{O}_2}$. The white dashed lines on each plot show the predicted desorption temperatures T_D of H_2O_2 from the three surfaces.

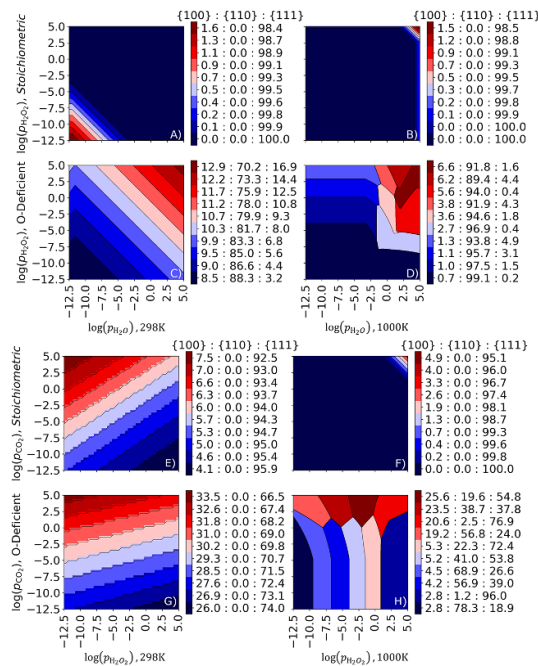
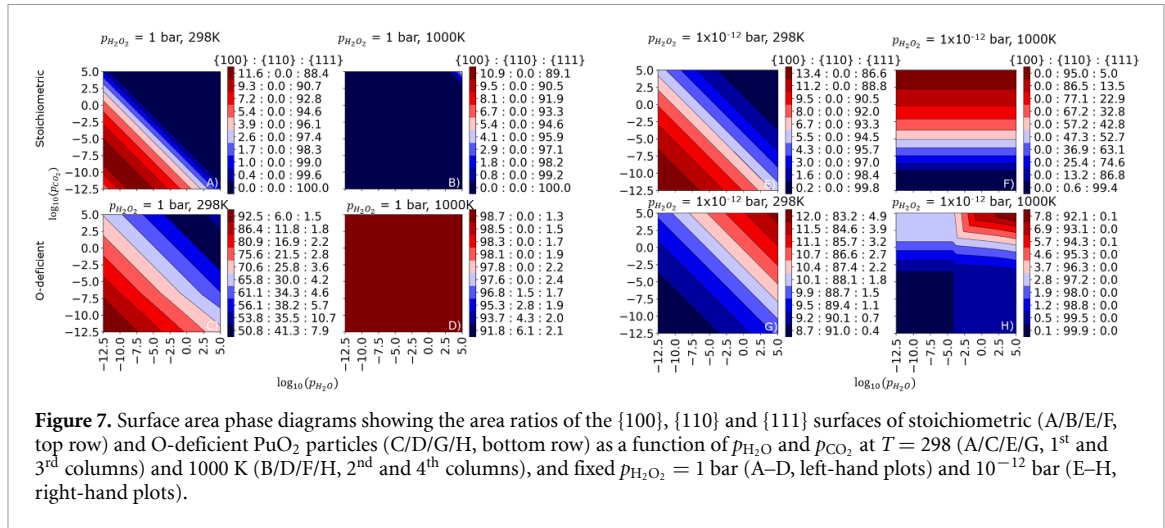


Figure 6. Surface area phase diagrams showing the area ratios of the {100}, {110} and {111} surfaces of stoichiometric (A/B/E/G, 1st and 3rd rows) and O-deficient PuO_2 particles (C/D/G/H, 2nd and 4th rows) as a function of $p_{\text{H}_2\text{O}}$ and $p_{\text{H}_2\text{O}_2}$ (A–D) and $p_{\text{H}_2\text{O}_2}$ and p_{CO_2} (E–H), at $T = 298$ (A/B/E/G, left column) and $T = 1000$ K (B/D/F/H, right column).

{100} varying from 4–7.5% at 298 K over a wide range of $p_{\text{H}_2\text{O}_2}$ and p_{CO_2} , and from 0–5% at 1000 K at high p_i . On the other hand, O-deficient PuO_2 particles are predicted to show markedly different behaviour. While we predict dominant {111} expression for O-deficient particles in the presence of H_2O_2 and CO_2 at 298 K, we predict 25–35% expression of the {100}, which, as for stoichiometric particles, varies over a wide range of p_i . Moreover, at $T = 1000$ K, we predict that O-deficient particles in the presence of H_2O_2 and CO_2 can express all three major surfaces. At p_{CO_2} up to approx. 1 bar, the proportion of the {110} and {111} decrease and increase, respectively, with $p_{\text{H}_2\text{O}_2}$. Above this threshold p_{CO_2} , different $p_{\text{H}_2\text{O}_2}$ result in the expression of all three surfaces in different proportions, with up to 25.6% {100}, 56.8% {110} and 76.9% {111}. For O-deficient particles in the presence of H_2O and H_2O_2 , we predict dominant {110} expression, with up to 16.9 and 6.6% of the other two major surfaces accessible at 298 and 1000 K, respectively, over a wide range of $p_{\text{H}_2\text{O}}$ and $p_{\text{H}_2\text{O}_2}$.

Analysing the effect of three co-adsorbates on the surface expression using similar 2D plots is more complex and requires fixing the temperature and the p_i of one of the three adsorbates. Figure 7 shows predicted phase diagrams as a function of $p_{\text{H}_2\text{O}}$ and p_{CO_2} , for stoichiometric and O-deficient particles, temperatures of 298 and 1000 K, and reference high and low $p_{\text{H}_2\text{O}_2} = 1$ and 10^{-12} bar. A full set of diagrams, including equivalent analyses where the $p_{\text{H}_2\text{O}}$ and p_{CO_2} are fixed, are provided in figure S6.

For stoichiometric particles, we generally predict dominant expression of the {111} surface, with 10–20% of the {100} accessible under some conditions. As mentioned previously, this can possibly be understood as balancing the lower surface energy of the {111} against the generally stronger binding of adsorbates at the {100} (figure 1). We observe one notable deviation from this trend in our predictions, namely that in the presence of low $p_{\text{H}_2\text{O}_2} = 10^{-12}$ bar at 1000 K we predict that the surface expression can be varied from nearly 100% {111} to 100% {110} by increasing the p_{CO_2} , regardless



of the $p_{\text{H}_2\text{O}}$ (figure 7(F)). This result is not obtained without H_2O_2 present [30], or at higher $p_{\text{H}_2\text{O}_2}$ (cf Figure 7(B)), and highlights the potential impact of the complex interactions between multiple adsorbates and the PuO_2 surface. The more similar γ_{Bare} of the O-deficient {110} and {111}, and the stronger affinity for adsorbates at these surfaces, means that the surface expression of O-deficient PuO_2 particles is predicted to be much more sensitive to the adsorbate partial pressures. In most cases, we predict dominant {110} expression, with significant and in some cases exclusive expression of the {100}, e.g. in the presence of H_2O and CO_2 at a high $p_{\text{H}_2\text{O}_2} = 1$ bar (figure 7(D)), which makes the large proportion of the {111} predicted under some conditions in figures 6(G) and (H) appear to be outliers.

A general observation from this study is that raising the temperature in the presence of H_2O_2 tends to shift the surface expression towards the dominant {111} for stoichiometric particles, and the {110} for O-deficient particles. The behaviour of the stoichiometric particles is easily explained by the fact that 1000 K is sufficient to desorb almost all adsorbates except at high p_i (figures 4 and S4), and the differences in surface energy favour the {111}. This trend also extends to H_2O (cf Figure 5) and holds in the presence of CO_2 and/or H_2O_2 [30]. The behaviour of O-deficient particles is more nuanced, because the {110} and {111} surfaces are closer in surface energy, the T_D are generally higher and in many cases above 1000 K, and the E_{Ads} of some (co-)adsorbed species strongly favours one surface (figure 1).

On this point, we found in previous work that presence of CO_2 and H_2O favours a mix of the {100} and {111}, which is more in keeping with the behaviour in the presence of CO_2 and H_2O_2 in figure 6, although these calculations also predicted that raising the temperature from $T = 298$ to 1000 K increases the expression of the {110}, especially at low p_i .

3.4. Particle morphology

Finally, the predicted surface compositions can be used to predict the shapes of PuO_2 nanoparticles (PuNPs) using the Wulff construction [62], which provides a means to link our atom-scale predictions to nanoscale experimental observables. In general, the particle shape determines facet-dependant properties including stability and reactivity. The particle morphology can in principle be controlled during preparation, and for the isostructural CeO_2 it is possible to kinetically stabilise non-equilibrium morphologies under some conditions [63–74].

However, if PuNPs are stored under incompatible environmental conditions, they will experience a thermodynamic driving force for reconstruction and morphological changes during ageing. Given the long timescales for PuO_2 storage, this possibility needs to be factored into the selection of appropriate disposal waste forms and storage conditions. Our model provides a simple but powerful means to approach this by allowing the impact of environmental conditions including temperature, oxygen chemical potential and adsorbate partial pressures on the thermodynamically favoured (equilibrium) particle morphology to be predicted.

Here, we investigate the morphologies accessible to stoichiometric and O-deficient particles at $T = 298$ or 1000 K, in the presence of H_2O_2 , $\text{H}_2\text{O}_2/\text{CO}_2$, $\text{H}_2\text{O}/\text{H}_2\text{O}_2$ and $\text{H}_2\text{O}/\text{H}_2\text{O}_2/\text{CO}_2$ from $p_i = 10^{-12}$ –1 bar. The predictions can be visualised in the form of ternary phase diagrams where the three vertices represent PuNPs expressing one of the major {100}, {110} or {111} surfaces, the edges represent particles expressing two surfaces, and the interior represents particles expressing all three surfaces.

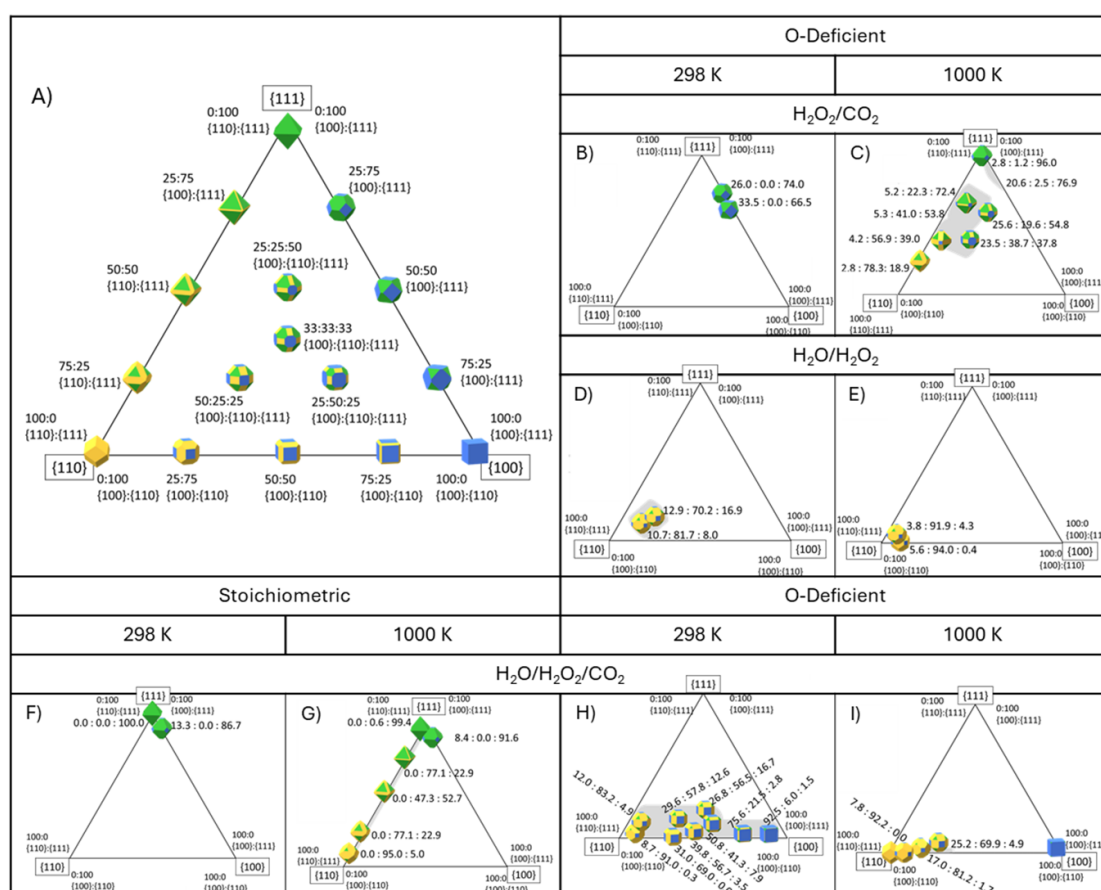


Figure 8. Ternary ‘morphology diagrams’ showing the predicted particle morphology of PuO_2 nanoparticles as a function of the relative areas of the $\{111\}$ (green), $\{110\}$ (yellow) and $\{100\}$ (blue) surfaces. The three vertices represent particles expressing a single surface, the three edges represent particles expressing two surfaces, and the interior represents particles expressing all three surfaces. (A) Range of possible particle morphologies assuming all surface area ratios are accessible. (B–I) Accessible particle morphologies for stoichiometric (F/G) and oxygen deficient particles (B/C/D/E/H/I) at $T = 298$ (B/D/F/H) and 1000 K (C/E/G/I), in the presence of co-adsorbed $\text{H}_2\text{O}_2/\text{CO}_2$ (B/C), $\text{H}_2\text{O}/\text{H}_2\text{O}_2$ (D/E), and $\text{H}_2\text{O}/\text{H}_2\text{O}_2/\text{CO}_2$ (F/G/H/I) at partial pressures between 10^{-12} and 1 bar. In (B–I) the accessible morphological space is marked by the shaded grey area. The particle shapes were generated using the Wulff construction, as implemented in the WulffPack Python Package [75] and used through pymatgen [47]. (A) Reproduced from [30]. CC BY 4.0. (B) Reproduced from [54]. CC BY 3.0.

Figure 8(A) demonstrates the particle shapes obtained when all possible ratios of the three surfaces are accessible. As demonstrated in the previous section, different combinations of environmental conditions will limit the surface expression and produce ‘morphology gaps’, where certain particle shapes are not thermodynamically stable and are unlikely to be accessible except under non-equilibrium conditions. Selected morphology diagrams are shown in figure 8, and a full set are provided in figure S7. To aid the discussion, we also summarise in table 3 the particle shapes we predict are accessible for stoichiometric and O-deficient PuNPs at 298 and 1000 K in the presence of different (co-)adsorbates at fixed low and high $p = 10^{-12}$ and 1 bar.

In the presence of H_2O_2 , we predict that both stoichiometric and O-deficient PuNPs are dominated by the $\{111\}$ surface and adopt an octahedral morphology (figure S7). At 1000 K, we predict that O-deficient PuNPs can express up to 17.1% $\{110\}$ and form truncated octahedral shapes located along one edge of the morphology diagram. This shape is obtained at high temperature and low $p_{\text{H}_2\text{O}_2}$ ($T = 1000$ K and $p_{\text{H}_2\text{O}_2} = 10^{-12}$ bar in table 3). We predict that the exposed surfaces of the stoichiometric octahedral PuNPs will be bare, whereas those of the O-deficient octahedral PuNPs will have adsorbed H_2O_2 under most conditions except at high T and low $p_{\text{H}_2\text{O}_2}$. This reflects the stronger affinity of the O-deficient surfaces for H_2O_2 and other adsorbates (cf figure 1). We predict that the truncated octahedral O-deficient PuNPs will have bare surfaces under the conditions at which they are formed, which is noteworthy because it implies that the presence of H_2O_2 has a sufficient impact on the surface energies even above the predicted desorption temperature.

As found in the previous section, the particle shapes accessible to stoichiometric PuNPs in the presence of additional adsorbates are relatively consistent. We predict the same dominant octahedral

Table 3. Summary of the energetically-accessible shapes for stoichiometric and O-deficient PuO_2 nanoparticles in the presence of different adsorbates at temperatures $T = 298$ or 1000 K and adsorbate pressures $p_i = 10^{-12}$ or 1 bar. For each entry we report the ratio between the $\{100\}:\{110\}:\{111\}$ surfaces and the corresponding surface speciation. A cross (X) denotes that a given particle shape is not accessible under a given set of conditions. We note that we only consider conditions where all adsorbates are present at the same p_i , and thus the full range of shapes indicated as accessible in figure 8 may not be listed in this table.

	Adsorbed species: H_2O_2			
	$T = 298$ K		$T = 1000$ K	
	$p_i = 10^{-12}$ bar	$p_i = 1$ bar	$p_i = 10^{-12}$ bar	$p_i = 1$ bar
Stoichiometric				
Octahedral	0 : 0 : 100 - : - : B	0.3 : 0 : 99.7 H_2O_2 : - : H_2O_2	0 : 0 : 100 - : - : B	0 : 0 : 100 - : - : B
O-deficient				
Octahedral	0 : 0 : 100 - : - : H_2O_2	0 : 0 : 100 - : - : H_2O_2	X	0 : 0 : 100 - : - : H_2O_2
Truncated octahedral	X	X	0.4 : 17.1 : 82.5 B : B : B	X
	Adsorbed species $\text{CO}_2/\text{H}_2\text{O}_2$			
	$T = 298$ K		$T = 1000$ K	
	$p_i = 10^{-12}$ bar	$p_i = 1$ bar	$p_i = 10^{-12}$ bar	$p_i = 1$ bar
Stoichiometric				
Octahedral	X	X	0 : 0 : 100 - : - : B	0 : 0 : 100 - : - : B
Truncated octahedral	5.3 : 0 : 94.7 $\text{H}_2\text{O}_2/\text{CO}_2$: - : B	5.7 : 0 : 94.3 $\text{H}_2\text{O}_2/\text{CO}_2$: - : $\text{H}_2\text{O}_2/\text{CO}_2$	X	X
O-deficient				
Truncated octahedral	27.6 : 0 : 72.4 $\text{H}_2\text{O}_2/\text{CO}_2$: - : H_2O_2	31.0 : 0 : 69.0 $\text{H}_2\text{O}_2/\text{CO}_2$: - : $\text{H}_2\text{O}_2/\text{CO}_2$	X	X
Rhombicuboctahedral	X	X	2.8 : 78.3 : 18.9 B : B : B	X
Octahedral (flat edges/vertices)	X	X	X	5.3 : 22.3 : 72.4 $\text{H}_2\text{O}_2/\text{CO}_2$: H_2O_2 : H_2O_2

(Continued.)

Table 3. (Continued.)

Adsorbed species: H ₂ O ₂				
	T = 298 K		T = 1000 K	
	$p_i = 10^{-12}$ bar	$p_i = 1$ bar	$p_i = 10^{-12}$ bar	$p_i = 1$ bar
Stoichiometric				
Adsorbed species H ₂ O/H ₂ O ₂				
	T = 298 K		T = 1000 K	
	$p_i = 10^{-12}$ bar	$p_i = 1$ bar	$p_i = 10^{-12}$ bar	$p_i = 1$ bar
Stoichiometric				
Octahedral	X	0 : 0 : 100 - : - : H ₂ O/H ₂ O ₂	0 : 0 : 100 - : - : B	0 : 0 : 100 - : - : B
Truncated octahedral	1.6 : 0 : 98.4 B : - : H ₂ O	X	X	X
	T = 298 K		T = 1000 K	
	$p_i = 10^{-12}$ bar	$p_i = 1$ bar	$p_i = 10^{-12}$ bar	$p_i = 1$ bar
O-deficient				
Rhombicuboctahedral	8.5 : 88.3 : 3.2 H ₂ O/H ₂ O ₂ : H ₂ O ₂ : H ₂ O ₂	11.2 : 78.0 : 10.8 H ₂ O/H ₂ O ₂ : H ₂ O/H ₂ O ₂ : H ₂ O/H ₂ O ₂	0.7 : 99.1 : 0.2 B : B : B	3.6 : 94.6 : 1.8 H ₂ O ₂ : H ₂ O ₂ : H ₂ O ₂
Adsorbed species CO ₂ /H ₂ O/H ₂ O ₂				
	T = 298 K		T = 1000 K	
	$p_i = 10^{-12}$ bar	$p_i = 1$ bar	$p_i = 10^{-12}$ bar	$p_i = 1$ bar
Stoichiometric				
Octahedral	0 : 0 : 100 - : - : B	0 : 0 : 100 - : - : CO ₂ /H ₂ O 0 : 0 : 100 - : - : CO ₂ /H ₂ O	0 : 0 : 100 - : - : B	0 : 0 : 100 - : - : B 0 : 0 : 100 - : - : B

(Continued.)

Table 3. (Continued.)

Truncated octahedral	13.4 : 0 : 86.6 CO ₂ /H ₂ O : - : B 5.8 : 0 : 94.2 CO ₂ /H ₂ O : - : B 1.6 : 0 : 98.4 CO ₂ /H ₂ O : - : B	X	X	0.4 : 0 : 99.6 B : - : B
Octahedral (flat edges)	X	X	0 : 0.6 : 99.4 - : - : B	X
		$T = 298\text{ K}$		$T = 1000\text{ K}$
O-deficient	$p_i = 10^{-12}\text{ bar}$	$p_i = 1\text{ bar}$	$p_i = 10^{-12}\text{ bar}$	$p_i = 1\text{ bar}$
Truncated cuboctahedral	20.9 : 69.8 : 9.3 All : All : H ₂ O ₂ 8.5 : 88.2 : 3.3 All : All : H ₂ O ₂	26.3 : 73.7 : 0.0 All : All : -	0.1 : 99.9 : 0 B : B : - 0.8 : 98.8 : 0.3 B : B : - 0.7 : 99.2 : 0.1 B : B : -	10.3 : 86.4 : 3.3 H ₂ O ₂ /CO ₂ : H ₂ O ₂ : H ₂ O ₂ 10.8 : 89.2 : 0.0 H ₂ O ₂ /CO ₂ : H ₂ O ₂ : -
Truncated octahedral	X	X	X	X
Truncated cuboctahedral	X	45.8 : 47.6 : 6.5 All : All : All	X	X
Cuboctahedral	X	X	X	98.7 : 0 : 1.3 H ₂ O ₂ /CO ₂ : H ₂ O ₂ : H ₂ O ₂
Rhombicuboctahedral	8.7 : 91.0 : 0.4 All : All : H ₂ O ₂	56.1 : 38.2 : 5.7 All : All : All	X	X

morphology under most conditions with CO_2 and/or H_2O present in addition to H_2O_2 , with a small degree of truncation possible due to limited expression of the $\{100\}$ (figures 8(F/G) and S7). The presence of all three adsorbates notably allows for most shapes along the $\{111\}$ – $\{110\}$ edge of the morphology diagram to be accessed at 1000 K, but the dominant $\{110\}$ expression is only possible at low $p_{\text{H}_2\text{O}_2}$ and high p_{CO_2} and the corresponding particle shape is therefore not listed in table 3. Whereas we predict bare particles at 298 K in the presence of $p_{\text{H}_2\text{O}_2}$ at low partial pressure, we predict that co-adsorption favours the binding of one or more adsorbates to some of the exposed surfaces and in particular the $\{100\}$. We consistently predict bare surfaces at the higher $T = 1000$ K, however, even at high p_i .

O-deficient PuNPs in the presence of different combinations of adsorbates can access a much wider range of particle shapes. Whereas we predict octahedral or truncated octahedral particles in the presence of H_2O_2 , we predict that the presence of H_2O_2 and CO_2 favours particles expressing a mix of the $\{100\}$ and $\{111\}$ at low temperature and provides access to a wide variety of morphologies at $T = 1000$ K (figure 8(B/C)). In the presence of H_2O_2 and H_2O , on the other hand, we predict particle shapes close to the $\{110\}$ vertex (figure 8(D/E)). In the presence of all three species, we predict morphologies from along the $\{100\}$ – $\{110\}$ edge of the morphology diagram, with up to 30% expression of the $\{111\}$ possible at low temperature (figure 8(H/I)). The more negative adsorption energies and higher desorption temperatures at the O-deficient surfaces mean that O-deficient PuNPs are invariably predicted to have bound adsorbates at low temperature (table 3). The propensity for H_2O_2 to heal the surface V_O means that H_2O_2 is always adsorbed, and in most cases we predict the other adsorbate(s) are also present. At $T = 1000$ K and low $p_i = 10^{-12}$ bar, we consistently predict bare surfaces, whereas high $p_i = 1$ lead to adsorption of H_2O_2 and often co-adsorbed CO_2 when present, but usually not H_2O . It is again notable that the presence of some adsorbates is predicted to lead to very different particle morphologies even when the temperature and p_i favour bare surfaces. This is well illustrated by the contrast between the octahedral particles expressing predominantly the $\{111\}$ surface predicted in the presence of H_2O_2 at $T = 1000$ K and $p_{\text{H}_2\text{O}_2} = 10^{-12}$ bar, and particles expressing dominant $\{110\}$, with very different morphology, under the same conditions with co-adsorbates present. This again highlights that the presence of multiple adsorbates can significantly influence the equilibrium particle morphology even when present at low partial pressure and not bound to the surface with appreciable coverage.

4. Conclusions

Control over the particle morphology is crucial in ensuring the long-term stability of stored PuO_2 powders. Various particle shapes and ordered/disordered aggregates can be formed depending on the source of the Pu [76] and preparation conditions [77]. Ensuring that the powders are stable under the planned storage conditions, and anticipating changes due to ageing in the presence of environmental contaminants, will allow for the reactivity to be predicted and the impact of undesirable chemical processes such as corrosion to be mitigated. The predictions from the present work clearly highlight the complexity of this problem: the presence of multiple adsorbates can lead to complex interactions, including both synergistic and antagonistic effects on the adsorption energies, and can lead to significant changes in the energetically accessible particle morphologies even when adsorbates are present at low partial pressures.

We can nonetheless attempt to draw out some general trends. The most important is that the behaviour of stoichiometric PuNPs appears to be relatively predictable, with octahedral particles dominated by the $\{111\}$ surface favoured under a wide range of conditions and a tendency for bare surfaces at high temperature. This suggests that controlling the oxygen partial pressure and harnessing the natural decay heat of the Pu could help to limit surface reactions and suppress radiolytic corrosion. However, this likely needs to be set against the possibility of combustion in oxygen-rich atmospheres. On this point, the one example in this study where we predict the preference for octahedral particles does not hold is at high partial pressures of CO_2 in the presence of low partial pressures of H_2O and H_2O_2 , the latter of which could be generated from self-radiolysis, and thus the use of CO_2 as a fire suppressant could have unexpected consequences.

The similar surface energies of the O-deficient $\{110\}$ and $\{111\}$, in conjunction with the stronger adsorbate binding and the larger impact of co-adsorption on the energetics, make the behaviour of O-deficient particles much less predictable. The results from this study suggest the equilibrium particle morphologies are highly sensitive to the presence of ubiquitous environmental contaminants. The generally stronger binding also means particles are likely to have surface-bound adsorbates, even at high temperatures, with a particularly strong affinity for species such as H_2O and H_2O_2 that can heal surface oxygen vacancies. This means that the potential for PuO_2 powders to be exposed to reducing atmospheres is likely to be of particular concern for ageing during long-term storage.

The modelling approach demonstrated in this work provides a simple but powerful means to predict the impact of different environmental conditions on PuO₂ particle morphology and surface speciation, and could therefore be a valuable predictive tool for identifying suitable disposal waste forms and storage conditions. While a number of limitations to the present model were highlighted during the discussion, a strength of our approach is that its accuracy can be systematically improved to account for additional variables. For example, an updated model could include additional surface configurations to account for the presence of more (co-)adsorbates, the interactions between adsorbates, and different levels of surface coverage. A related improvement would be to explicitly account for the O₂ partial pressure in our model, rather than distinguishing between the behaviour of stoichiometric and O-deficient surfaces with a fixed level of O deficiency as in the present work. Improvements could also be made to the underlying theory, principally a more accurate descriptions of the energetics with dispersion-corrected DFT, including solvation effects, and temperature corrections accounting for the contribution of vibrations to the free energy.

Ultimately, the predictions from the model will need to be verified against experimental measurements, which is itself challenging due to the difficulties inherent in working with PuO₂. To support this, we envisage an iterative process where a modelling approach similar to the present work is used to help prioritise experiments, and data from the experiments is then used to improve the model. If specific adsorption configurations of interest are identified, it would also be possible to analyse the relevant models in more detail to obtain microscopic insight into the interactions between the adsorbates and surfaces e.g. by analysing the electronic density of states or atomic charges. Our models could also form the basis for exploring surface reactivity. In our view, the results from this study therefore highlight the clear value of computational modelling for addressing the challenging issue of managing and disposing of legacy Pu stockpiles, and we will continue to develop the approach in future work.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.17632/sddw6xwwc5> [48].

Supporting information available at <http://doi.org/10.1088/1361-648X/ae35b6/data1>.

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Author contributions

Conceptualization: SCP, MM; Data curation: SM, MM; Formal Analysis: SM, JMF, JMS, MM; Funding Acquisition: LJG, DJC, MM; Investigation: SM, LJG, DJC, MM; Methodology: SM, JMS, MM; Project Administration: MM; Resources: LJG, DJC, MM; Supervision: LJG, DJC, MM; Validation: SM, JMS, MM; Visualization: SM, JMS, LJG, DJC, SCP, MM; Writing—Original Draft: SM, JMF, JMS, MM; Writing—Review & Editing: SM, JMF, JMS, LJG, DJC, SCP, MM.

Conflict of interest

There are no conflicts to declare.

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